

Task 19 — Inventory Turnover Analysis

Professional stock-movement and slow-mover analysis using Excel and SQL.

Rows	76,000	Annualized Turnover	10,776.55x
Categories	5	Slow Movers	16
Stockout Rate	0.53%	Top Category	Groceries

1. Objective

Analyze stock movement, compare product categories, and identify slow-moving products. The analysis follows the task deliverables: a turnover table and a slow-mover list.

2. Methodology

The supplied dataset does not contain COGS. Traditional financial inventory turnover therefore cannot be calculated directly. For this project, a units-based turnover measure is used: $\text{Turnover Ratio} = \text{Total Units Sold} \div \text{Average Inventory}$; $\text{Annualized Turnover} = \text{Turnover Ratio} \div \text{observation years}$. This is explicitly labeled as a stock-movement metric rather than financial inventory turnover.

Slow movers are the bottom 25% of Product ID + Category combinations by annualized turnover. The bottom 10% are labeled Critical Slow Mover. Thresholds: Q1 = 98.28x; P10 = 71.29x.

3. Category Turnover Results

Category	Units Sold	Avg Inv.	Annualized Turnover	Stockout Rate
Groceries	3,127,335	360.3	4,171.28x	27.37%
Clothing	1,150,873	282.1	1,960.63x	7.24%
Toys	834,679	229.1	1,750.85x	6.71%
Furniture	880,654	250.1	1,692.43x	6.05%
Electronics	757,335	289.3	1,258.28x	6.05%

Finding: Groceries has the highest stock-movement turnover at 4,171.28x. Electronics has the lowest category turnover at 1,258.28x.

4. Slow-Mover Watchlist

Product	Category	Class	Turnover	Units Sold	Avg Inv.
P0005	Electronics	Critical Slow Mover	61.62x	64,867	505.9
P0017	Furniture	Critical Slow Mover	63.31x	48,824	370.6
P0005	Toys	Critical Slow Mover	63.56x	61,601	465.8

Product	Category	Class	Turnover	Units Sold	Avg Inv.
P0010	Furniture	Critical Slow Mover	67.49x	49,267	350.8
P0014	Electronics	Critical Slow Mover	67.66x	61,899	439.6
P0003	Clothing	Critical Slow Mover	69.69x	82,721	570.4
P0011	Clothing	Critical Slow Mover	71.02x	73,231	495.5
P0015	Clothing	Slow Mover	71.90x	71,402	477.3
P0020	Clothing	Slow Mover	73.30x	73,031	478.9
P0015	Furniture	Slow Mover	79.29x	49,667	301.0
P0007	Toys	Slow Mover	80.39x	61,349	366.8
P0003	Furniture	Slow Mover	80.46x	51,036	304.8
P0016	Toys	Slow Mover	87.04x	61,518	339.7
P0002	Electronics	Slow Mover	93.74x	62,916	322.6
P0016	Furniture	Slow Mover	94.13x	47,259	241.3

Recommended action: review reorder levels, demand patterns, seasonality and promotional opportunities before carrying excess stock. Critical slow movers should be prioritized for management review.

5. Interview Takeaways

What is inventory turnover? It measures how quickly inventory is sold and replenished. Financially, it is COGS divided by average inventory.

Why can high turnover be risky? If replenishment is too slow, inventory may become too lean, causing stockouts and lost sales. High turnover should therefore be assessed alongside stockout rate and service level.

6. Deliverables

Excel workbook: dashboard, category turnover table, product turnover table, slow-mover watchlist, monthly trend, data quality, methodology and raw data. SQL file: PostgreSQL queries for data quality, turnover, category comparison, slow-mover identification and monthly trend.